

# Ruşen Birben

AI Researcher & Engineer

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AI and Data Engineering graduate from Istanbul Technical University and independent researcher with METU's Applied NLP Group. Research interests center on LLM reliability and interpretability, with experience in Turkish NLP, machine unlearning, and LLM applications for finance.

## EDUCATION

### B.Sc. AI and Data Engineering

2022 - 2025

Istanbul Technical University

GPA: 3.61 / 4.00

Graduated at 09.07.2025

### Electronics and Communication Engineering

2020 - 2022

Istanbul Technical University

Transferred to AI & Data Engineering

## SCHOLARSHIPS & AWARDS

### 1416 YLSY Scholarship · M.Sc. & Ph.D.

2026

Awarded government funding for overseas master's study followed by doctoral study, covering tuition fees, a monthly living stipend, health insurance, travel, language-exam fees, and study supplies. Following completion of the funded studies, will return to Türkiye as an assistant professor.

### Kaggle Bronze Medal · The Pokémon Company PTCG AI Battle Challenge

2026

Ranked 677th of 6,807 teams (top 10%) in Kaggle's simulation competition, developing and evaluating partially observable game-playing agents with replay-based behavioral cloning, belief-aware Monte Carlo planning, and self-play.

## RESEARCH EXPERIENCE

### Independent Researcher

03/2026 - Present

METU Applied NLP Group

Ankara, TR

- Contribute to data creation and evaluation methodology for the SIGTURK 2027 benchmark on Turkic languages and cultural understanding with Çağrı Toraman.
- Support LlamaTurk2 through training-data collection and pretraining experiments.
- Lead METU NLP Connect, producing expert interviews.

### AI Ops Research Intern

08/2024 - 09/2024

Havelsan

Istanbul, TR

- Investigated software-log analysis for predictive maintenance; developed prototypes using HDFS logs.

### NLP Research Intern

07/2024 - 08/2024

ITU NLP Lab

Istanbul, TR

- Researched financial LLM applications involving sentiment analysis, tool use, reasoning, forecasting, and retrieval-augmented generation. Contributed experiments to the Kaira undergraduate capstone.

## PUBLICATIONS & RESEARCH CONTRIBUTIONS

**Exact First, Then Fast: An Event-for-Event C++ Rebuild of the Agenton ABIDES Market** 2026  
*Agenton Workshop @ NeurIPS 2026 - Accepted for in-person poster presentation* [🔗](#) Competition

- Göktürk Batın Dervişoğlu, Muhammed Faruk Aytin, Muhammed Rüsen Birben.
- Rebuilt the Agenton ABIDES market simulator in C++, preserving event ordering, random-number sequences, and reference outputs. Validated exact agreement across 6,900 randomized comparisons and achieved a 1,299× simulation-loop speedup on the largest public benchmark, measured on the same laptop.
- Atlanta, USA, December 12, 2026.

**Turkish Cultural Knowledge Dataset** 2026  
*MRL Workshop @ EMNLP 2026 - Accepted dataset contribution; joint benchmark paper in preparation*  
[🔗](#) Shared Task

- Led the source-grounded agent pipeline and data preparation for a Turkish cultural knowledge dataset spanning 15 domains.
- Screened 1,200 candidate question-answer pairs; selected 500 for sociological review by Büşra Tufan and prepared the final 300-pair submission.
- Contributed to English translation review under the research guidance of Çağrı Toraman.

## SELECTED PROJECTS

**Kaira** 2025  
*Undergraduate Capstone: Financial Reasoning LLM*

- Fine-tuned Qwen-based LLMs with GRPO, Unsloth, and LoRA; ran FinQA and GSM8K experiments on reward design and financial reasoning. Developed an Ollama-based chat application.

**Tobor** 2024 - 2025  
*Autonomous Warehouse Robot* [🔗](#) Code [📺](#) Video

- Implemented the box pickup/drop-off mechanism in a team-built ROS2 warehouse robot with navigation and YOLO perception; shared data annotation and supported other algorithms.

**Alzheimer** 2024  
*Machine Unlearning in Stable Diffusion* [🔗](#) Code

- Adapted Forget-Me-Not-based unlearning for Stable Diffusion 2.1. Implemented attention controllers and explored loss objectives; qualitatively assessed concept removal and retention of unrelated generation capabilities.

**Biting The Bytes** 2024  
*Turkish Diacritic Restoration* [🔗](#) Code [🔗](#) Demo

- Led ByT5-small training and hyperparameter tuning with PEFT/LoRA for byte-level Turkish diacritic restoration. Shared debugging and Hugging Face Spaces deployment with a teammate.

**To-AI-or-Not-to-AI** 2023  
*GPT Detector* [🔗](#) Code [🔗](#) Demo

- Combined three fine-tuned RoBERTa detectors by majority vote. Shared preparation of 26,270 examples; achieved 94.21% accuracy on an in-domain random split, without establishing out-of-domain generalization.

## PROFESSIONAL EXPERIENCE

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### Co-Founder & Technical Lead

08/2025 - 04/2026

*Fiction Studios*

Ankara, TR

- Co-founded Fiction Studios at Bilkent Cyberpark; led technical development and coordinated three additional contributors. Built a multilingual LLM-powered museum and tourism guide, alongside customer and investor discussions.

### Junior AI Engineer

03/2025 - 07/2025

*CyberQuote*

Istanbul, TR

- Developed news-sentiment pipelines, internal financial LLM tools, web integrations, and a mobile voice assistant at a Phillip Capital subsidiary. Worked on retrieving current context from fragmented sources and PDF reports.

## TECHNICAL SKILLS

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**Languages** Python / JavaScript/TypeScript / SQL / C/C++

**Frameworks & Tools** PyTorch / TensorFlow / Transformers / PEFT / LoRA / Unsloth / Scikit-learn / Pandas / NumPy / ROS2 / YOLO

**Specialties** NLP / LLMs / RAG / Machine Unlearning / Model Evaluation

## COURSES / LANGUAGES

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**Deep Learning Specialization**

DeepLearning.AI · Coursera

**AWS Cloud Technical Essentials**

AWS · Coursera

 **Turkish** Native **English** TOEFL iBT 109/120 (November 8, 2025)

**Interests:** Coffee / Books / Photography / Anime & Manga